

NLP/S&DAI

TakeThemOut: A dialogue driven stealth game

Edoardo Vassallo



**Università
di Genova**

DIBRIS DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

Creative Project

Difficulty: Not Specified

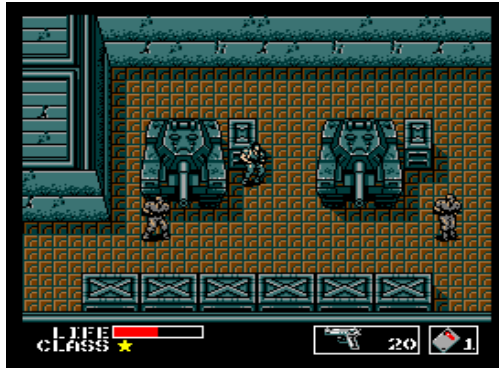


Proposal Summary:

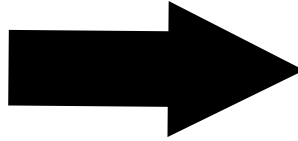
Development of a stealth game which makes use of VesNA-Pro for agents' mind design and Intent-based chatbots for natural language interactions, to improve enemy instances' characterization and hence player's immersion

Project Motivation

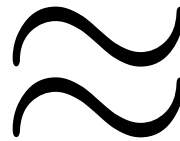
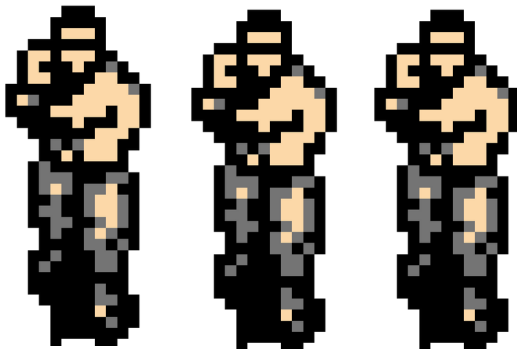
Uneven Evolution



Metal Gear (1987)



Metal Gear Solid Δ (2025)



How Do We Improve Instance Characterization?

Method 1: Temper Based Behaviour

If Shy:

Hide

...

If Aggressive:

Attack

...

If Scared:

Run Away

...

Else:

Default Behaviour

...

Method 2: Natural Language Interactions



Façade (2005)



**Università
di Genova**

DIBRIS DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

Project Overview

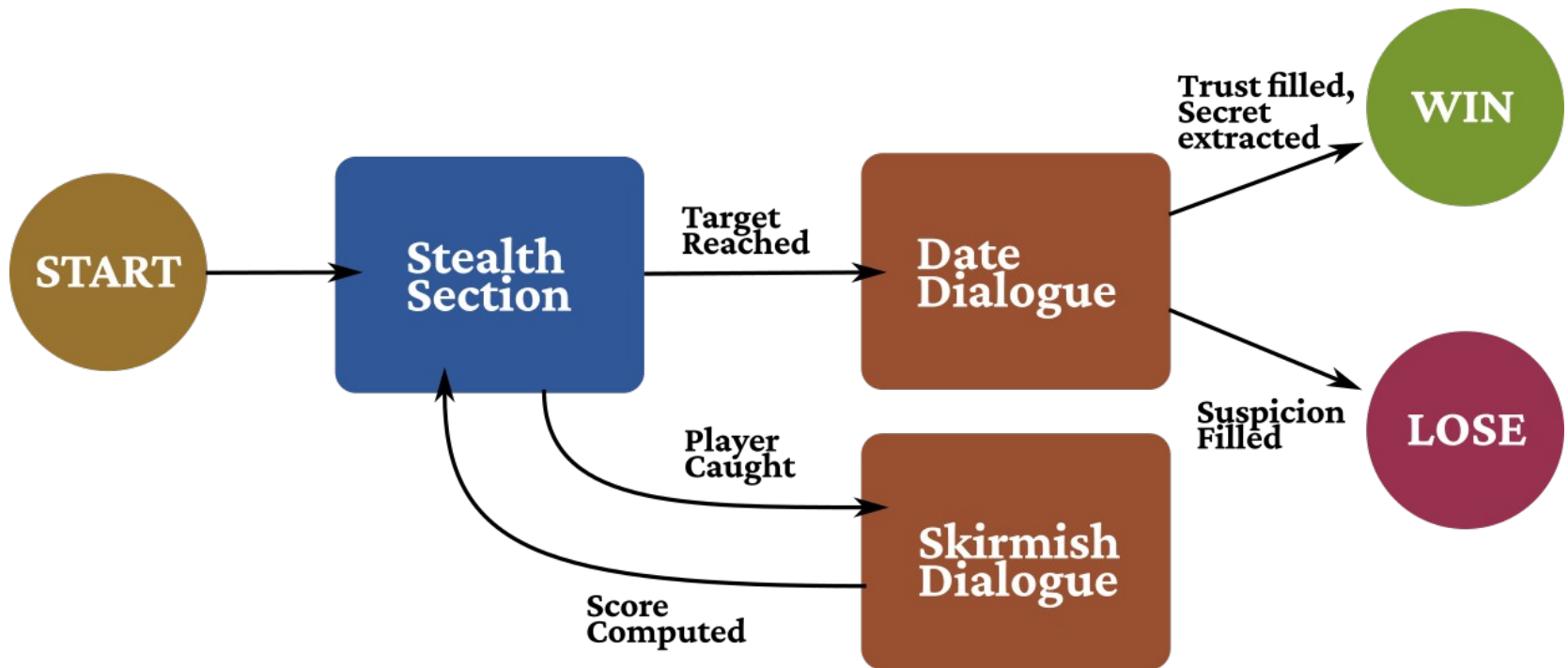
Technologies Used

Godot Engine – Environment and agent bodies design

VEsNA-Pro – Agents' minds design

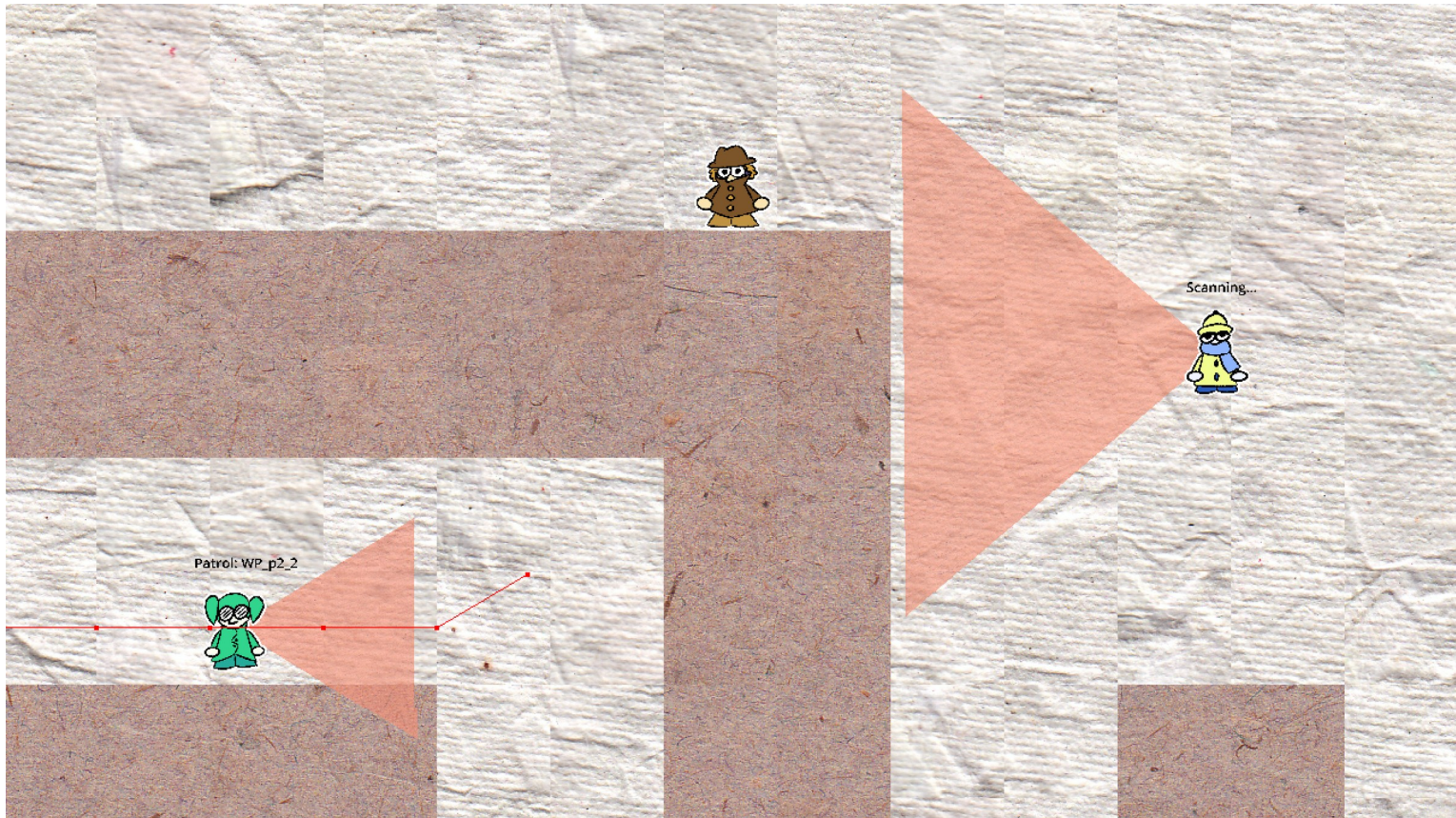
Rasa – Agents' chatbots design

Game Loop



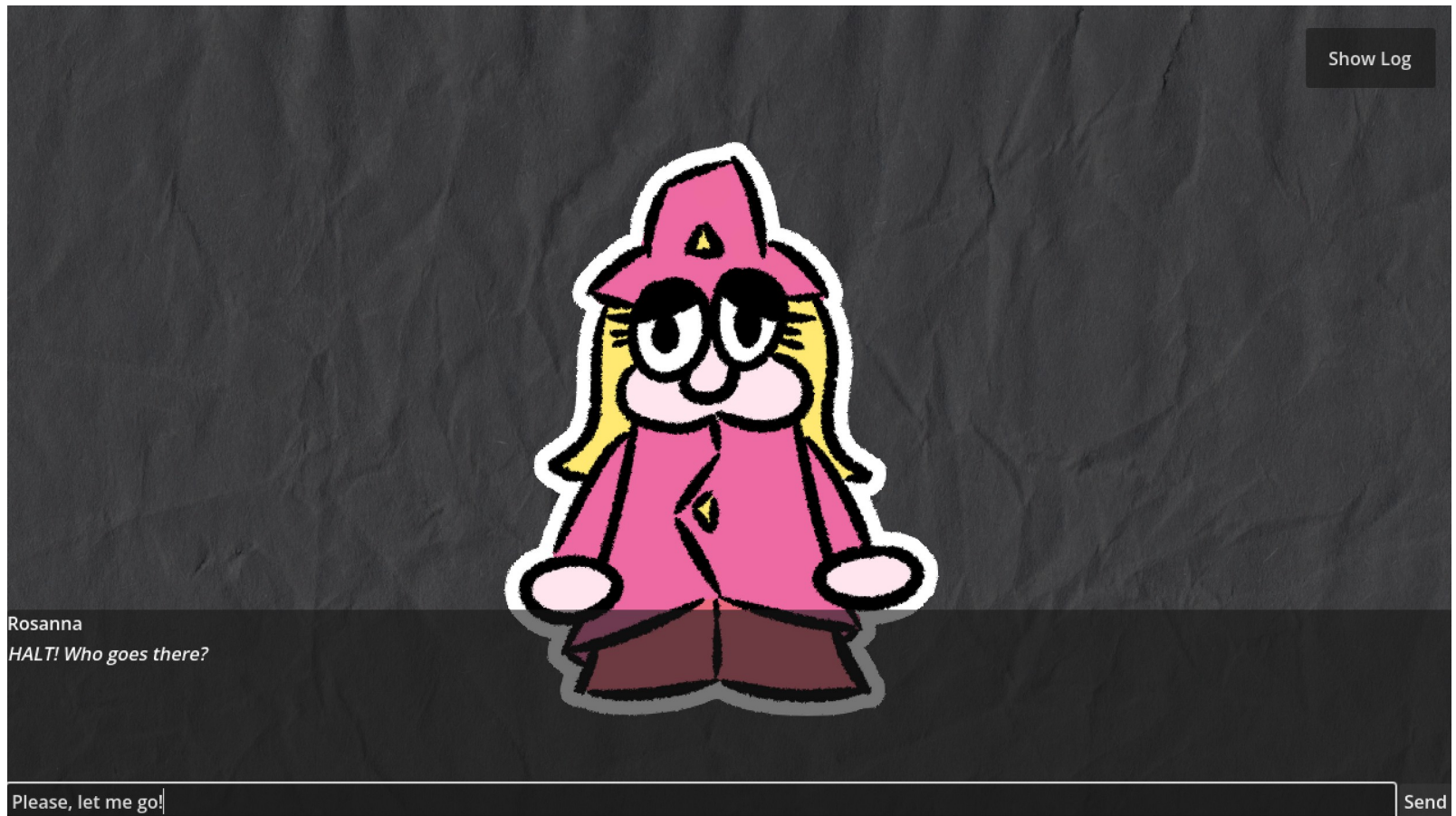
Schema detailing the different phases of the game, and the transitions between them.

Stealth Section



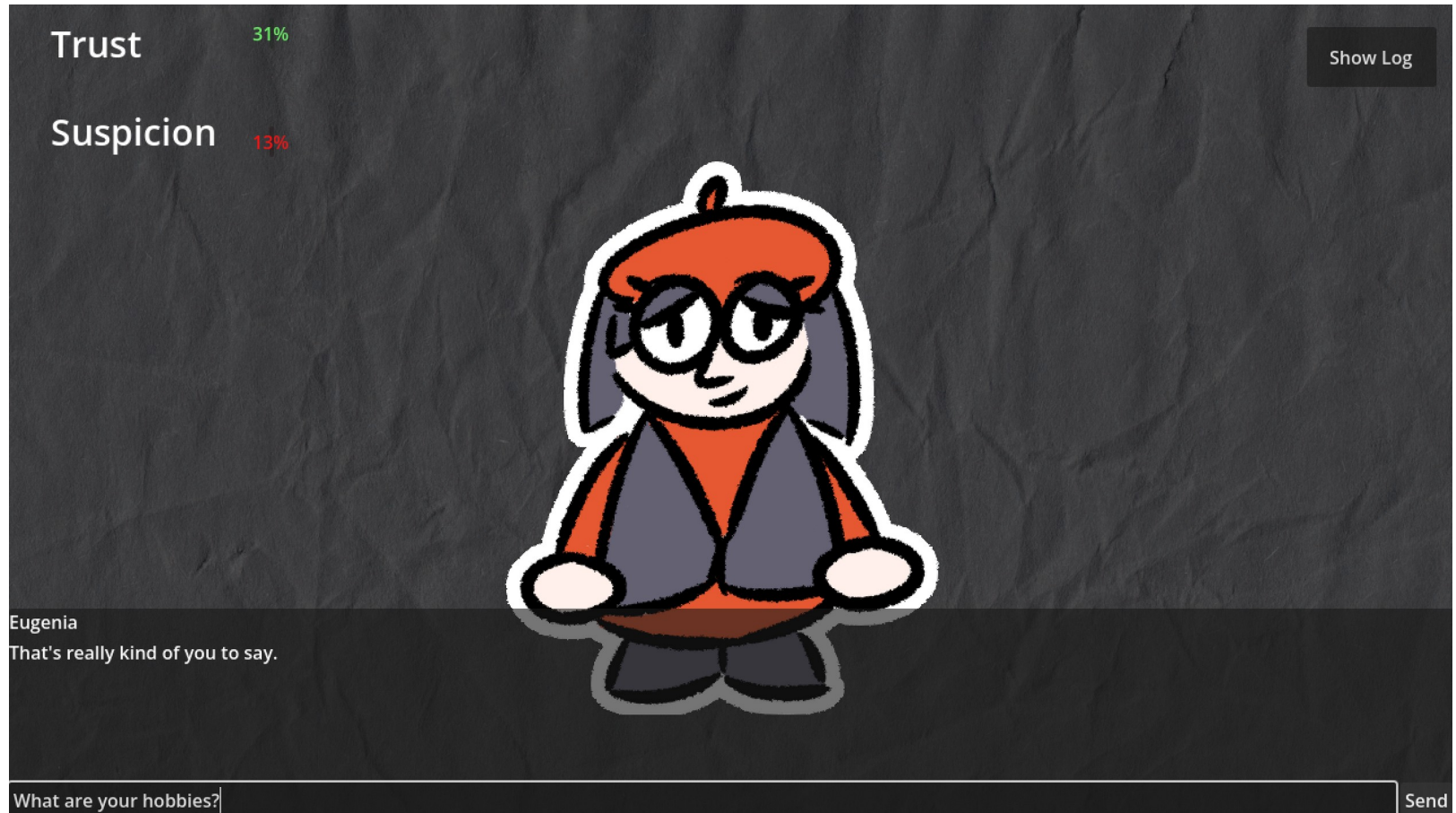
Screenshot of the Stealth Section

Skirmish Dialogue



Screenshot of the Skirmish Dialogue Section

Date Dialogue



Screenshot of the Date Dialogue Section



**Università
di Genova**

DIBRIS DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

Reasoning Design

Mind-Body Relationship

The bodies of the agents have been designed as **State Machines**

The mind is able to transition between the body's states and modify directly its parameters. Communication between agents happens at the mind's level

When met with complex events, the body notifies the mind and moves to an Idle state, waiting for its input.

Why Choose A State Machine Approach?

Agent Types

Three types of agents were developed:



Sentry



Patrol



Captain

Agent Types

Sentries

Sentries stand still and alternate between a set of view points. If they spot the player, they alert eventual allies nearby

Two body states:

- Scan
- Alert



Sentries

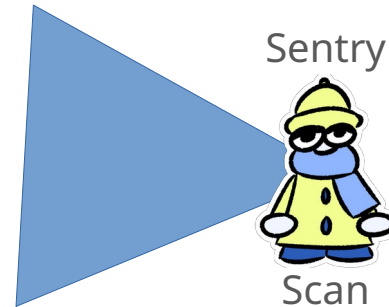
Example

The sentry starts in the Scan state, and alternates between two viewpoints

**Sentry
Mind**



Bodies



Sentries

Example

The sentry spots the player. The body notifies the mind and goes to the Idle state

Sentry Mind

Received
notification saying
Player has been
spotted!

Ally



Bodies

Sentry



Idle

Player



Sentries

Example

The mind sends the command to move to the Alert state. The body searches for allies nearby, and sends a list to the mind.

Sentry Mind

Send command to
move to Alert.

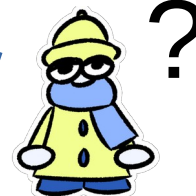
Received list of allies
nearby

Bodies

Ally



Sentry



Alert

Player



Sentries

Example

The mind notifies its allies. When the alert time is over, the body notifies the mind and moves to Idle.

Sentry Mind

Notify allies

Received
notification saying
Alert time is over.

Ally !



Bodies

Sentry



Idle

Player



Sentries

Example

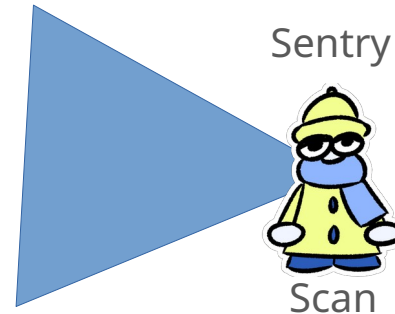
The mind can command to transition back to the Patrol State, closing the loop.

Sentry Mind

Send command to move to Scan.



Bodies



Agent Types

Patrols

Patrols move in the maze. When they see the player, they try to catch him, following him even after losing sight.

They answer to alerts from Sentries and Captains

Four body states:

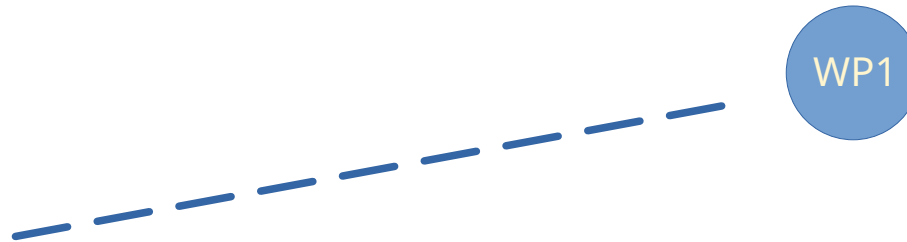
- Patrol
- Chase
- Track
- Investigate



Patrols

Patrol State

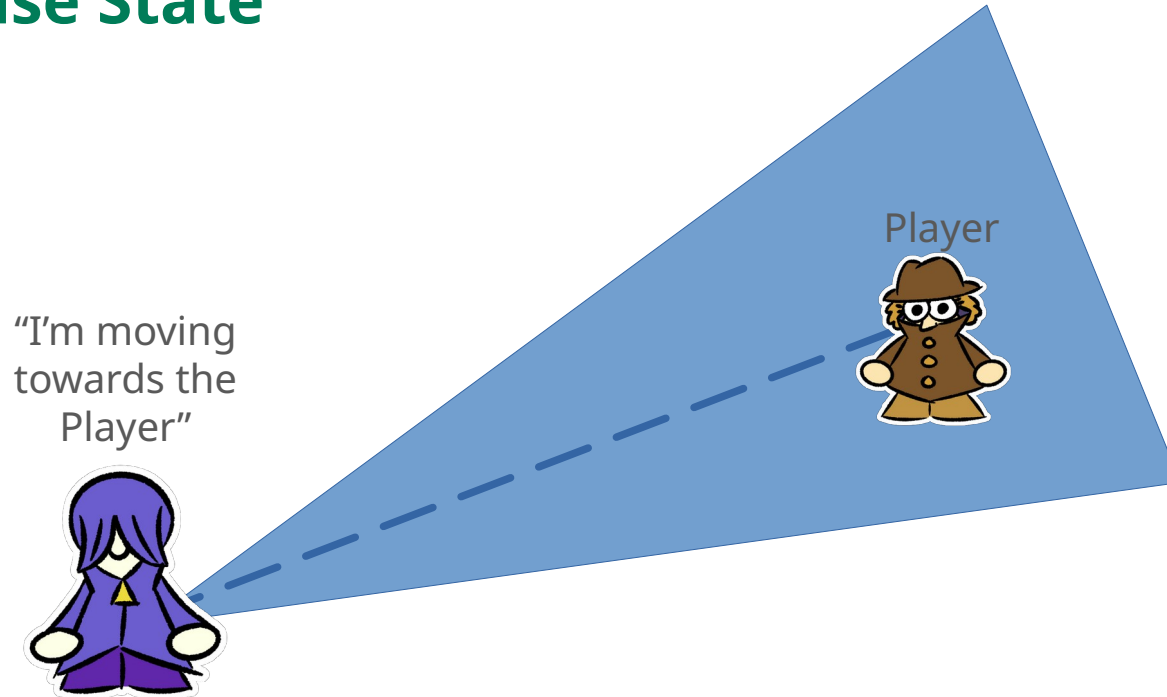
"I'm
navigating
to point
WP1"



Could also be a coordinate or Agent position

Patrols

Chase State



Patrols

Track State

"I'm
following
the player's
trail"



Player



Patrols

Investigate State

R3

"I'm patrolling
X random
points
nearby"



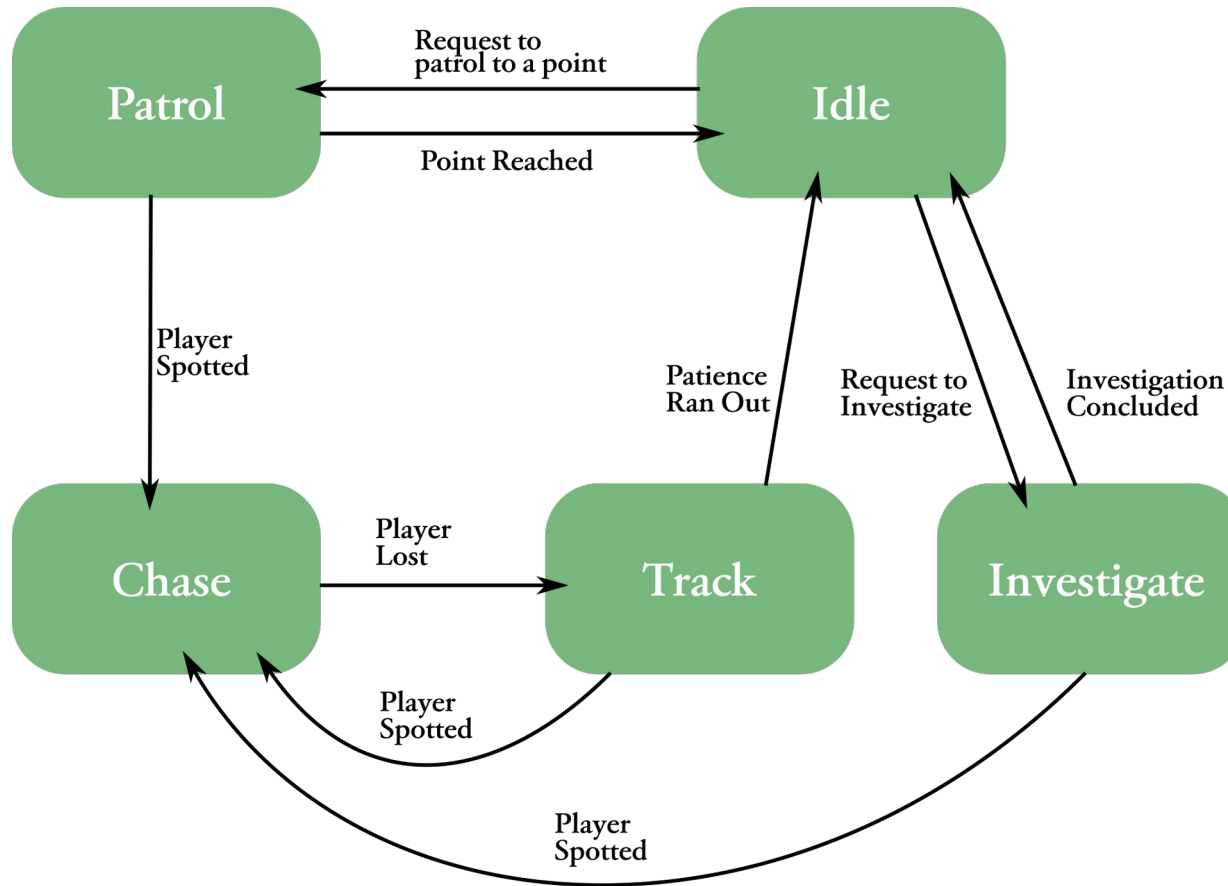
4

R1

R2

Patrols

Body Transitions



Patrols

Messenger System

"I'll follow the player,
go alert the nearest
captain"



Ally



Player



A "Super-State"
implemented on the
mind's level

Agent Types

Captains

Captains behave similarly to Patrols but add **coordination** and **communication**. They patrol between the last player's sighting locations and alert allies when they spot the player.



Captains

Patrol Computation

When deciding where to move next, the captain asks other agents for the last player sighting

Captain Mind

Ask allies for last
player sighting



Bodies



Captains

Patrol Computation

The allies reply. The captain will move to the **average** point reported.

Captain Mind

Received position L1
and L2

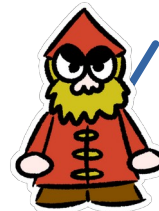
Send command to
Patrol to A1

Bodies

"I saw them in
L1"



L1



A1

"I saw them in
L2"

L2



Captains

Chase Alert

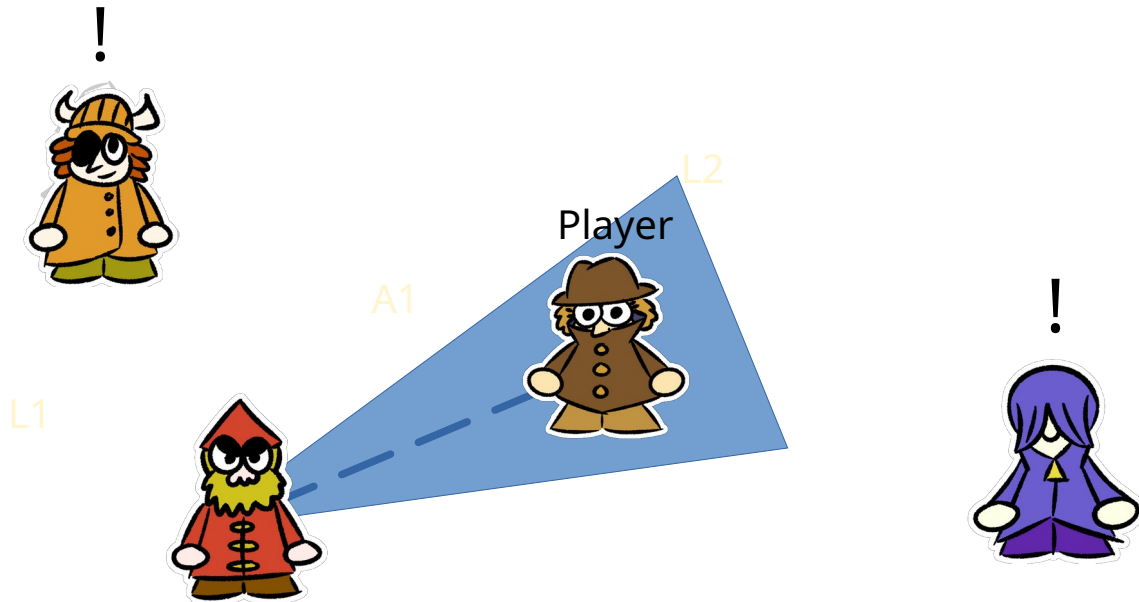
When the player is spotted, all available allies are alerted

Captain Mind

Received
notification saying
Player has been
spotted!

Alerting allies

Bodies





**Università
di Genova**

DIBRIS DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

Adding Temper

Temper Components

The minds do not choose their plans based only on beliefs and environment events, but also based on **personality** and **mood** traits

Traits

The two personality traits are:

- **Aggressiveness**
- **Laziness**

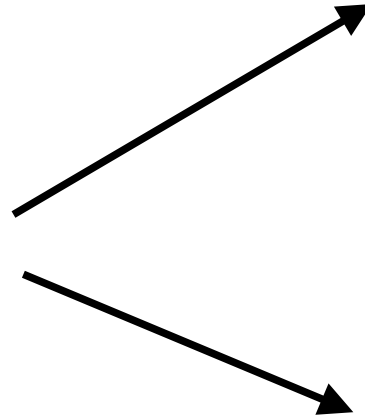
The two mood traits are:

- **Fear:** influenced by encounters with the player
- **Sympathy:** influenced by natural language interactions

Example

Sentry

Captain asks for
player's last sighting



Default plan:

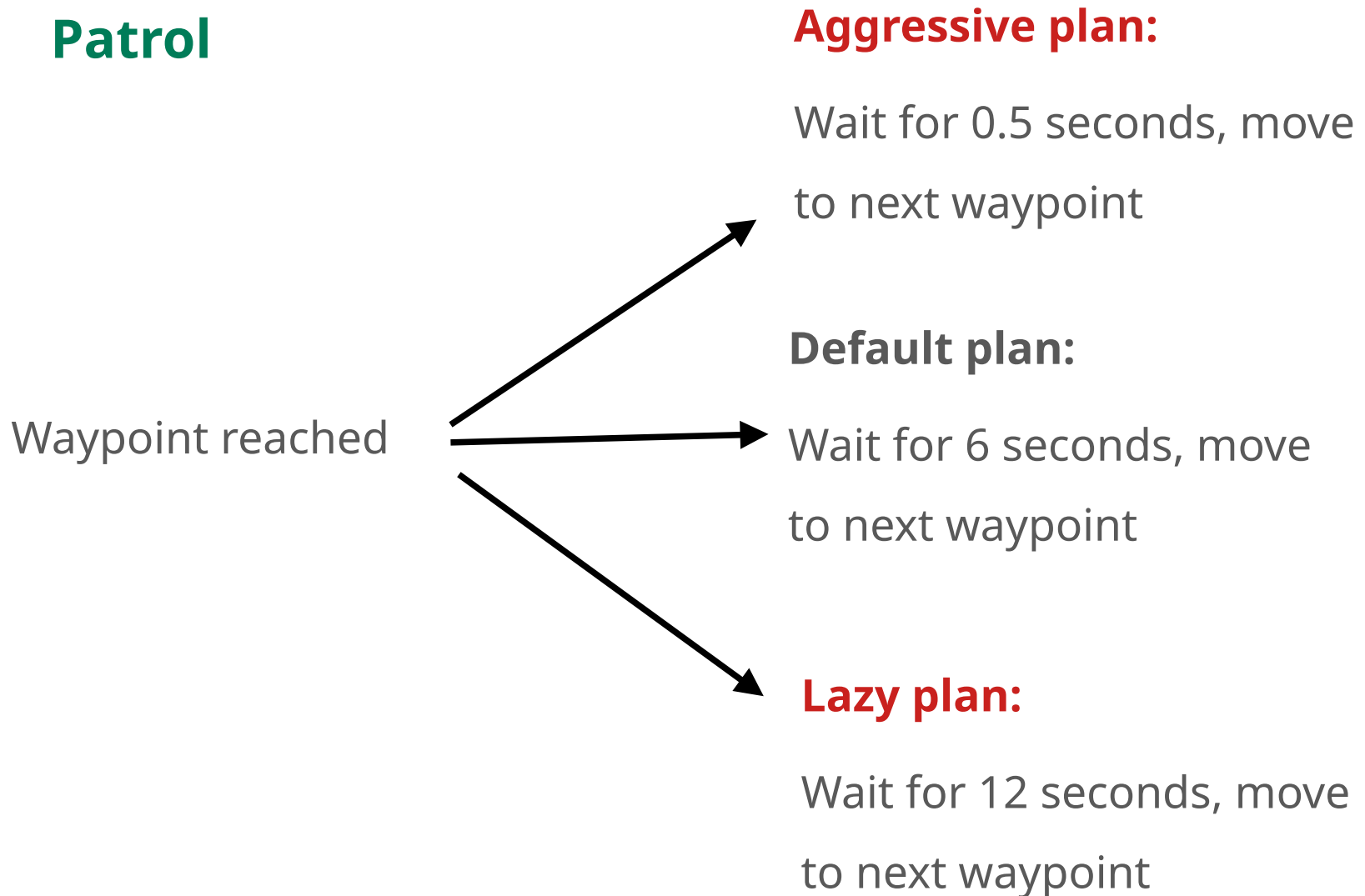
Send last recorded
position

Sympathetic plan:

Lie and send nothing

Example

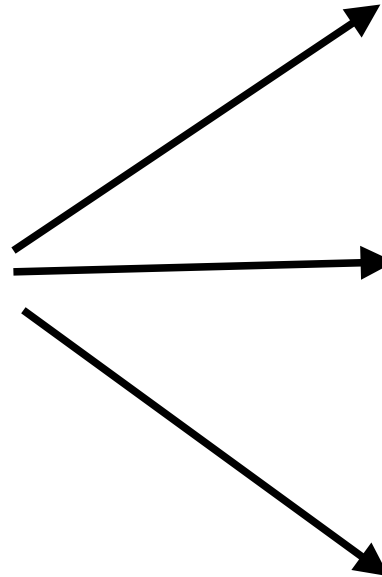
Patrol



Example

Captain

Player Spotted



Aggressive + Fear plan:

Do not alert allies, chase alone

Default plan:

Alert allies, start chasing the player

Lazy + Sympathetic plan:

Directly ignore the player



**Università
di Genova**

DIBRIS DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

Chatbot Design

Natural Language Interactions

The player is able to interact with a subset of the agents through natural language in the Skirmish and Date scenes

The agents' dialogue have been implemented through Rasa **intent-based** chatbots

Why Choose Intent-based Chatbots?

Natural Language Interactions

Two bots were developed, one for each scene, sharing the intent detection pipeline, but with different intents and action scripts

Skirmish Bot

Intents

The Skirmish Bot is able to detect the following intents:

- **Flattery**, example: *"You're clearly the best guard here"*
- **Aggression**, example: *"Don't you dare touch me"*
- **Logic**, example: *"My badge is valid for this sector"*
- **Pity**, example: *"Please have mercy on me"*
- **Bribe**, example: *"I can pay you off to look the other way"*

Skirmish Bot

Input Analysis

Based on the player's input, the chatbot first detects its **intent** and then its **sentiment** score using Python's VADER library.

Based on the current agent personality, a **sympathy score** is computed, and an answer is selected. The former is passed to the relative agent's mind

Skirmish Bot

Example Personalities

Instance	Flattery	Bribe	Logic	Aggression	Pity	Sentiment
Rosanna	+1.0	+0.5	+0.2	-1	-0.5	Positive
Susanna	+0.2	-0.5	+0.9	-1	+0.8	Positive
Polyanna	-0.8	-0.5	-0.2	+0.4	+0.7	Negative
Marianna	+0.6	+0.2	-0.6	-0.5	-0.8	Positive

$$s = \underbrace{w_i \cdot \delta_i}_{\text{intent}} + \underbrace{w_s \cdot (\tau \cdot d)}_{\text{sentiment}}$$

Date Bot

Intents

The Date Bot is able to detect the following intents:

- **Compliment**, example: *"I really like your dress"*
- **Curiosity**, example: *"What are your hobbies?"*
- **Talking About Oneself**, example: *"I'm thinking about changing careers"*
- **Toxicity**, example: *"I really can't stand this conversation"*
- **Asking the Objective**, example: *"Tell me the name of the target"*

Date Bot

Input Analysis

The bot keeps track of two values, a **Trust** score and a **Suspicion** Score. Based on the player's input and the previous two, the chatbot detects the current input and computes new updated scores.

If Suspicion maxes out, the player **loses** the game.

If Trust is high and Suspicion low enough, the player can *ask for the objective* and receive an answer, **winning** the game.

Date Bot

Example Scores

Intent	Cold Phase (T < 30)		Warm Phase (T < 50)		Close Phase (T > 50)	
	ΔT	ΔS	ΔT	ΔS	ΔT	ΔS
Compliment	+10	+13	+6	+11	+2	+9
Curiosity	+2	+3	+6	+3	+10	0
Talking about Self	+2	-3	+6	-6	+10	-3
Toxicity	-9	-2	-13	-5	-20	-9
Ask Objective	-15	+20	-9	+13	+2	+7

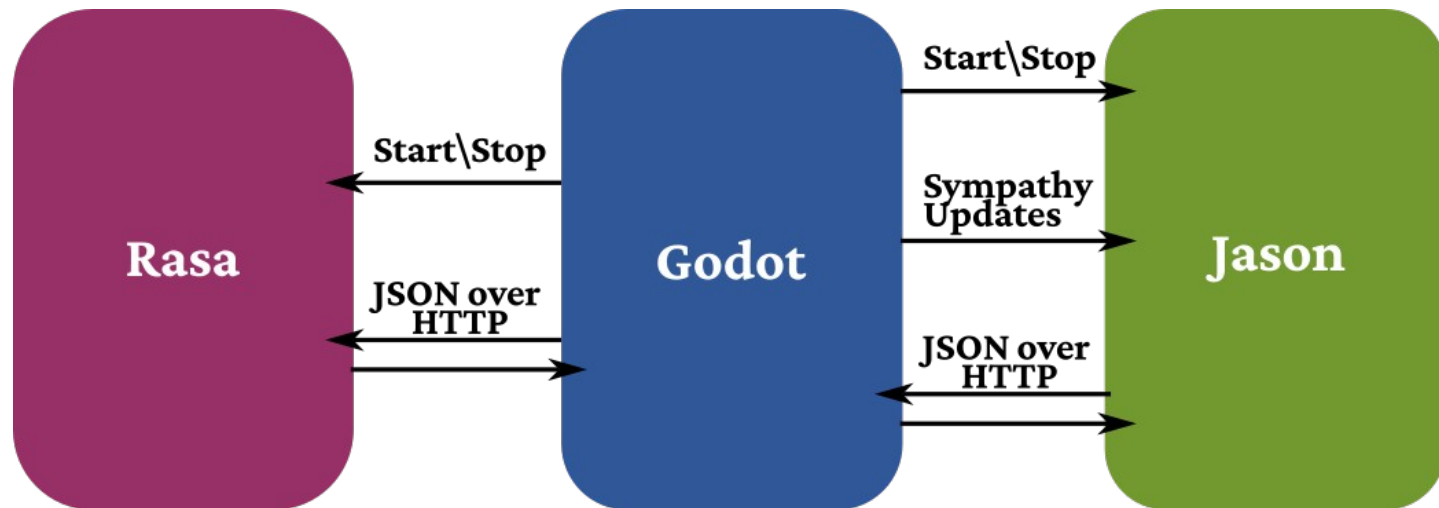


**Università
di Genova**

DIBRIS DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

Integration and Testing

System Integration



Simple schema summarizing communications between the main three system components

Testing

The system was developed on an Arch Linux machine, and tested on both Linux and Windows based machines.

Issues were found with initialization and startup of Jason and Rasa on the latter system.

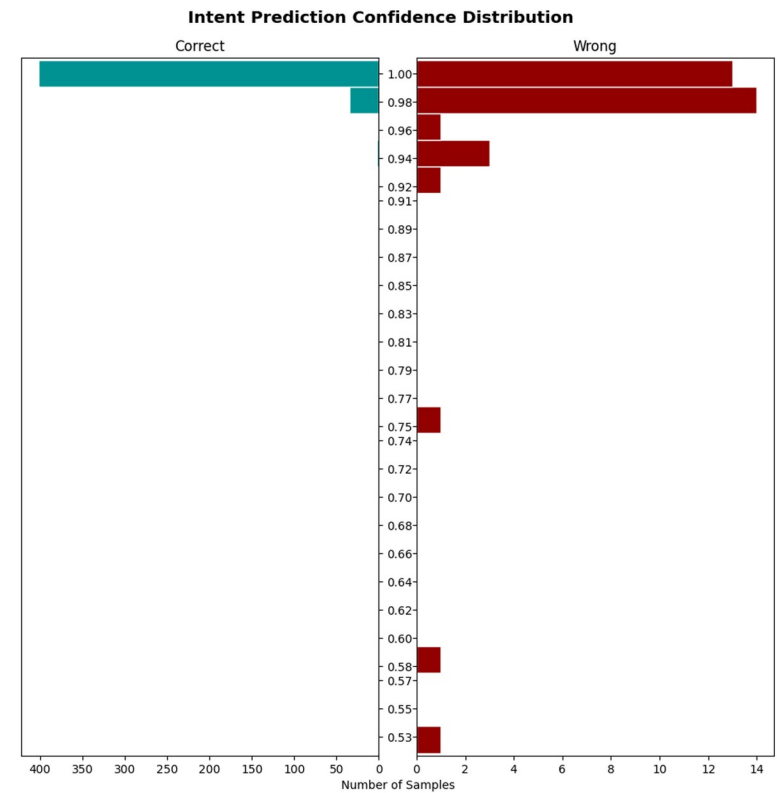
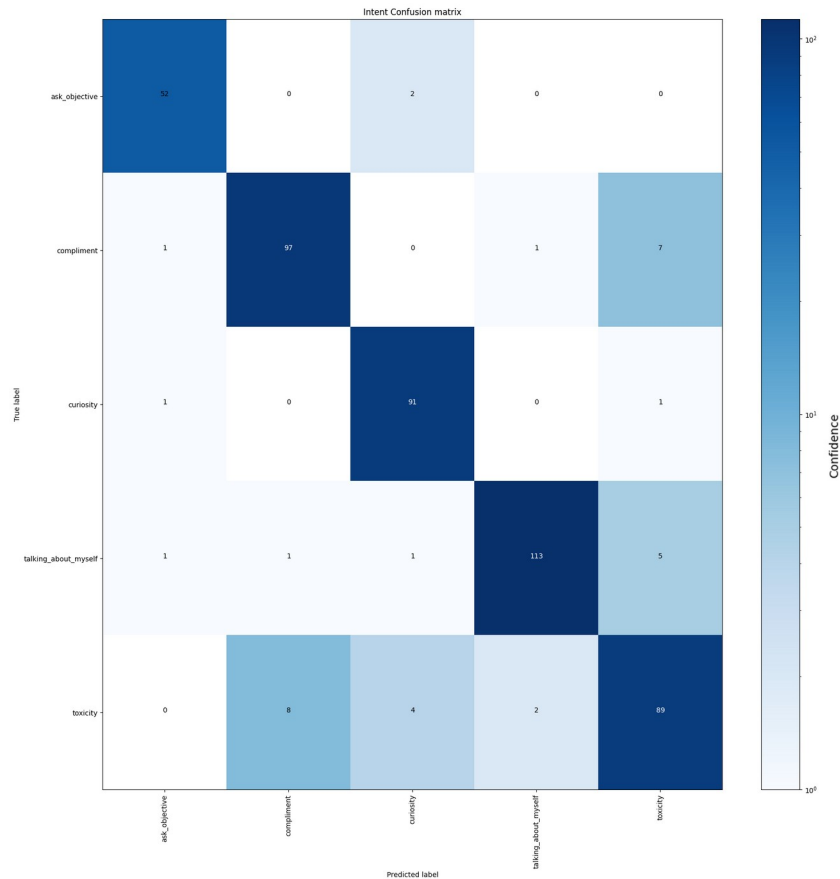
Issues

Besides the previous, the following issues were detected:

- Agent could often **get stuck** in the maze. This is a problem strictly related to Godot's navigation system
- Chatbots often **failed** to **generalize** and detect the correct intent on complex sentences. This is related to the quality and quantity of its training data.

Issues

Crossvalidation Results (Date chatbot)





**Università
di Genova**

DIBRIS DIPARTIMENTO
DI INFORMATICA, BIOINGEGNERIA,
ROBOTICA E INGEGNERIA DEI SISTEMI

Final Remarks

UniGe

DIBRIS